

Prime-Gap Memory and the Differencing Trap

The wobble looks 44% predictable; 42 of those points are an artifact — but the last 2 are the wheel

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What this is. A methodological result with a small genuine core. The eye-catching “prime weather” one sees in the *change* of the gaps — volatility clustering, big-jump recoil, a windowed wobble that looks nearly half-predictable — is **almost entirely an artifact of differencing**, reproduced by random data. The instrument that separates signal from artifact is a specific null: *shuffle the gaps, then apply the same transform*. Under that null a genuine residual survives — roughly **+2 points of R^2 in the windowed wobble**, about six times the memory in the raw gaps — and it is plausibly the wheel’s fingerprint on consecutive gaps. Tags: [emp] measured · [null] shuffle-control · [fact] exact · [interp] reading. Companion negative control: [ED_Negative_Control.md](#); frequency-space dual: [Prime_Structure_Factor.md](#).

All figures over the 269991 gaps between consecutive primes in $[10^6, 5 \times 10^6]$ (mean gap $\bar{g} = 14.81$, sd 12.29).

1. The seductive picture

Write $g_n = p_{n+1} - p_n$ for the gap and $\Delta g_n = g_{n+1} - g_n$ for the **jitter** (the change in gap, i.e. the second difference of the primes; up to the damping $1/p$ this is the angle-wobble $\Delta\alpha$ of the Prime-Triangle note). Plotted in order, the jitter looks alive:

- **Volatility clustering.** Big jitters arrive in bursts. The mean $|\Delta g|$ after a *large* jitter (top decile) is 22.0; after a *tiny* one, 10.7 — against an overall mean of 12.9. The autocorrelation of $|\Delta g|$ is +0.31.
- **Recoil.** After a large jitter, the next jitter has the *opposite* sign 81.8% of the time — the “spike and return.”
- **A predictable wobble.** Regress Δg_n on its last five values: the fit explains **43.7%** of the jitter variance.

Any one of these, reported alone, reads as a discovery about the primes. All three are mirages.

2. The instrument: shuffle, then transform

The primes’ program already carries the right reflex — the [ED negative control](#): a process that excludes *causally* rather than *arithmetically* diffracts as a liquid, proving the wheel’s crystal is arithmetic. The same move disciplines time-series claims. The null hypothesis is “*the gaps carry no ordering memory*” — so the null sample is the **same multiset of gaps in random order**. Crucially, any statistic computed through a transform must be compared to that transform **applied to the shuffled gaps**, not to a shuffle of the transformed series.

The trap, stated once. Shuffling Δg *directly* destroys the differencing structure you are trying to account for, and so credits the artifact to the primes. The jitter must be **re-formed from shuffled gaps**: $\Delta g^{\text{null}} = \text{diff}(\text{shuffle}(g))$.

Two nulls, two layers. Which thing you shuffle decides *which* structure you are testing, and the two are easy to conflate. Shuffling the **gaps** keeps them positive, so the reconstructed sequence still increases with bounded steps: the monotonic *skeleton* — and everything it forces, including the 45° angle ceiling and the tight change-cancellation of [The Prime-Triangle Angle](#) — **survives**; only the fine wheel *order* is destroyed. So **gap-shuffle tests “is it the wheel.”** To interrogate the skeleton itself you must shuffle the **changes** and let them free-float: that walk wanders straight through the ceiling (the angle crosses 45° and sags far below), which proves the cancellation is *not* a free property of the change-values but is held in place by the primes being a genuine monotonic, bounded-gap sequence. So **change-shuffle tests “is it the skeleton.”** Match the null to the layer, or a real structural fact reads as “generic noise.”

3. What the null reveals

statistic (window of 5, or lag 1)	real primes	correct null	genuinely prime
raw-gap memory —	0.32%	0.00%	+ 0.32 pp
regress g_n on last 5			
gap autocorrelation, lag 1	−0.046	−0.001	− 0.045
jitter clustering, $\text{corr}(\Delta g_n , \Delta g_{n-1})$	+0.315	+0.319	−0.004 (<i>artifact</i>)
jitter recoil, big $\rightarrow$ opposite sign	81.8%	79.2%	+2.7 pp
windowed wobble — regress Δg_n on last 5	43.67%	41.69%	+ 1.97 pp

Two clean verdicts:

- **The clustering and most of the recoil are artifact.** The null reproduces them to within a hair. They are not prime weather; they are what differencing does to *any* spread-out sequence (§5).
- **A genuine residual survives, and it concentrates in the window of the wobble.** The raw gaps carry +0.32 pp of memory; the windowed jitter carries +1.97 **pp** — six times more — none of it reproduced by the null. On 2.7×10^5 samples this is overwhelmingly significant.

The lesson for the whole gap program: the interesting quantity is not how predictable the wobble *looks* (44%) but how much of that survives the transform-matched null (+2%).

4. Why differencing manufactures the mirage

The artifacts are not subtle statistics; they are forced by algebra. For an i.i.d. sequence X with variance σ^2 , the first difference $Y_n = X_{n+1} - X_n$ has

$$\text{cov}(Y_n, Y_{n-1}) = \text{cov}(X_{n+1} - X_n, X_n - X_{n-1}) = -\text{Var}(X_n) = -\sigma^2, \quad \text{Var}(Y) = 2\sigma^2,$$

so $\boxed{\text{corr}(Y_n, Y_{n-1}) = -\frac{1}{2}}$ [**fact, exact**] — a hard −0.5 recoil, with *no* memory in X at all. That single identity is the “spike and return.” Likewise $|Y_n|$ and $|Y_{n-1}|$ share the term X_n : a large X_n inflates both, forcing a positive correlation of magnitudes — the “volatility clustering.” Regressing Y on its past then recovers the $-\frac{1}{2}$ structure and reports a large R^2 . All three headline effects are consequences of the shared term, present in the shuffled gaps and absent from the primes’ account.

5. What is genuinely prime — the wheel in the gap correlations

Strip the pedestal and the residual is real. Two shuffle-proof facts remain:

1. **Consecutive gaps are weakly anti-correlated**, $\text{corr}(g_n, g_{n+1}) = -0.046$ (null ≈ 0). A large gap is followed, a touch more often than chance, by a smaller one — a genuine reversion in the *levels*, not the differences.
2. **A window of a few gaps constrains the next**, +1.97 pp beyond the differencing null.

Both point at the same source. Consecutive gaps are not independent because the **wheel** forbids many gap-sequences outright and biases the rest: the residues mod 6, 30, 210 that one gap lands on shape which gaps can follow. (That the wheel *quantitatively* accounts for the level anti-correlation — a small-primes-only model reproduces the measured value with no residual — is verified non-circularly in [The Prime-Triangle Angle §4.1](#).) This is the gap-space face of the **Lemke Oliver–Soundararajan** correlations (the “unexpected biases in consecutive primes,” 2016), and the same coprimality template that casts every other shadow in

this collection. The memory in the wobble is small, but it is the wheel — read through the correlations of successive escapes.

6. Reading the record straight

This note corrects an over-eager reading (my own, made and caught while writing it): that the jitter’s clustering and recoil were prime “weather,” and that amplitude was the knowable structure while sign was noise. The null reverses both halves. The **amplitude** persistence is the artifact; the small **genuine** memory lives in the *ordering of the gap levels* and surfaces most in the windowed wobble. The honest one-line model of “how far to the next prime” is therefore: **the base rate $\ln p$ sets the scale, the wheel sets the admissible slots, and the last few gaps shift the odds by about two points of variance — no more.** Everything past that is the pseudo-random residual the structure factor already isolated as diffuse background; here it is the same residual in the time domain.

7. The key collapses to the wheel

If the surviving memory is that small, can it help you *find* the next prime? Build the predictor and measure it. From a prime p , the candidates are the **wheel-open slots** $p + g$ (positions coprime to the small primes); rank them by probability of being the next prime and test in that order. Three rankers, on 26,723 primes near 3×10^6 (trained on $[10^6, 3 \times 10^6]$):

ranker	mean tests to find	top-1	top-3
walk up the open slots	3.99	23.5%	56.8%
+ gap-frequency reweighting	3.99	23.5%	56.8%
+ the regime/history lever	3.99	23.5%	56.8%

The regime lever — given maximum generosity (the full next-gap distribution conditioned nonparametrically on the recent-regime tercile) — moves the mean rank by +0.0007. Gap-frequency reweighting never puts a different slot first than “walk up” (0.0% of cases). The reason is exact: to land on the k -th open slot, every earlier open slot must be composite, so $P(\text{next prime} = k\text{-th slot})$ **strictly decreases in k** — walking up the wheel is Bayes-optimal, and a frequency-trained ranker merely rediscovers it. The rank distribution is a clean geometric decay (23.5, 18.8, 14.5, 11.2, 8.1, ... %) — Cramér-on-the-wheel: given the wheel, each escape is memoryless.

So the whole “how to find the next prime” key collapses to one line: **go to the next wheel-open slot.** The wheel is 100% of the exploitable structure, the order within it is forced, and the genuine +2 pp of windowed memory — real as it is — sits below the threshold that could reorder even the first test. This is the sharpest form of the collection’s thesis: the order is entirely the wheel, and the residual is not merely hard to predict but **provably useless** for locating the next escape.

Appendix. Reproduction

```
import random
def sieve(n):
    b=bytearray([1])*(n+1); b[0]=b[1]=0; i=2
    while i*i<=n:
        if b[i]: b[i*i::i]=bytearray(len(b[i*i::i]))
        i+=1
    return b
B=sieve(8_000_000)
P=[i for i in range(1_000_000,5_000_000) if B[i]]
g=[P[i+1]-P[i] for i in range(len(P)-1)]
```

```

diff=lambda s:[s[i+1]-s[i] for i in range(len(s)-1)]

def R2(series,k=5):                                # in-sample R^2 predicting next from last k
    import numpy as np
    S=np.array(series,float)
    X=np.column_stack([S[k-1-j:len(S)-1-j] for j in range(k)]); y=S[k:]
    X=np.column_stack([X,np.ones(len(y))]); b,_,_=np.linalg.lstsq(X,y,rcond=None)
    return 1-((y-X@b)**2).sum()/((y-y.mean())**2).sum()

gs=g[:]; random.shuffle(gs)                        # shuffle the GAPS, then difference
print("levels   real",round(R2(g),4), "null",round(R2(gs),4))
print("wobble   real",round(R2(diff(g)),4),"null",round(R2(diff(gs)),4))
# key: the null jitters are diff(gs), NOT a shuffle of diff(g).

```

References

- A. Proxmire, *The Prime-Triangle Angle* (this collection) — the wobble $\Delta\alpha$, and the earlier runs/shuffle observation this note refines.
- A. Proxmire, *The ED Negative Control* and *The Prime Structure Factor* (this collection) — the shuffle/negative-control discipline, and the order/randomness split in frequency space.
- R. J. Lemke Oliver, K. Soundararajan, *Unexpected biases in the distribution of consecutive primes* (PNAS, 2016) — residue correlations of successive primes; the mechanism behind the surviving gap memory.